

EMNLP 2026

Oral

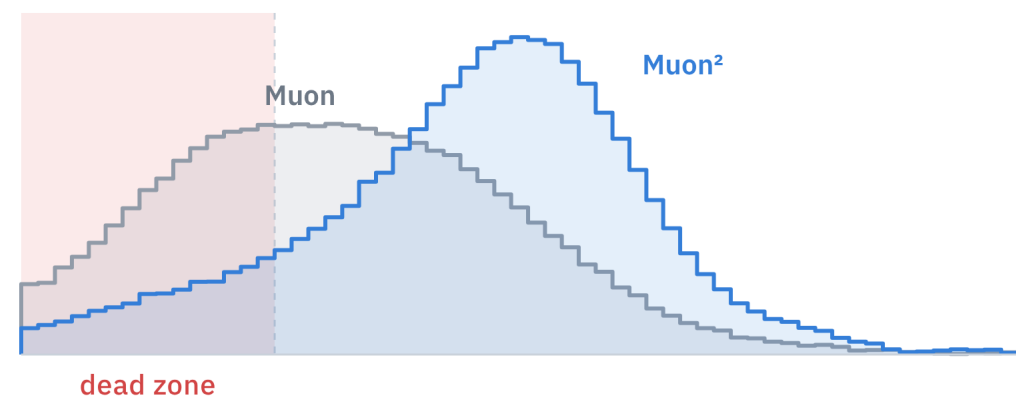
Budapest, Hungary

Muon²: Boosting Muon via Adaptive Second-Moment Preconditioning

Ziyue Liu¹, Ruijie Zhang¹, Zhengyang Wang¹, Yequan Zhao¹, Yupeng Su¹, Zi Yang², Zheng Zhang¹

¹University of California, Santa Barbara

²University at Albany, SUNY



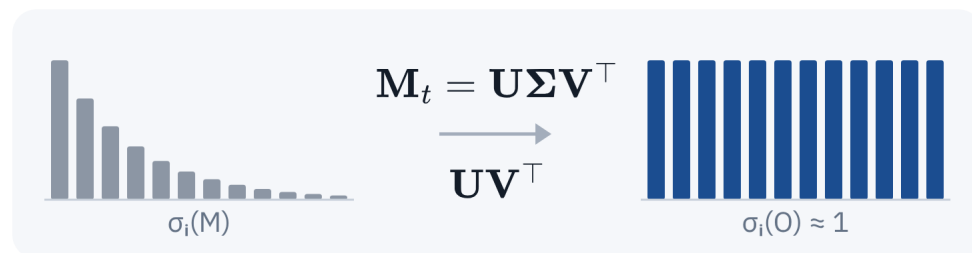
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Muon orthogonalizes the momentum of each weight matrix

Pre-training is expensive, and the optimizer decides how many steps a model needs.

Muon (Jordan et al., 2024) has become a strong optimizer for LLM pre-training.



Keep the singular directions, set **every singular value to 1**.

Exact SVD is too slow, so Muon runs **K** Newton–Schulz iterations ($K = 5$ by default).

USED TO PRE-TRAIN

Moonlight Liu et al., 2025

Kimi Kimi Team, 2025

GLM-4.5 Zeng et al., 2025

Fewer Newton–Schulz steps make Muon worse

ONE NEWTON–SCHULZ STEP

$$\mathbf{X} \leftarrow a\mathbf{X} + b(\mathbf{X}\mathbf{X}^\top)\mathbf{X} + c(\mathbf{X}\mathbf{X}^\top)^2\mathbf{X}$$

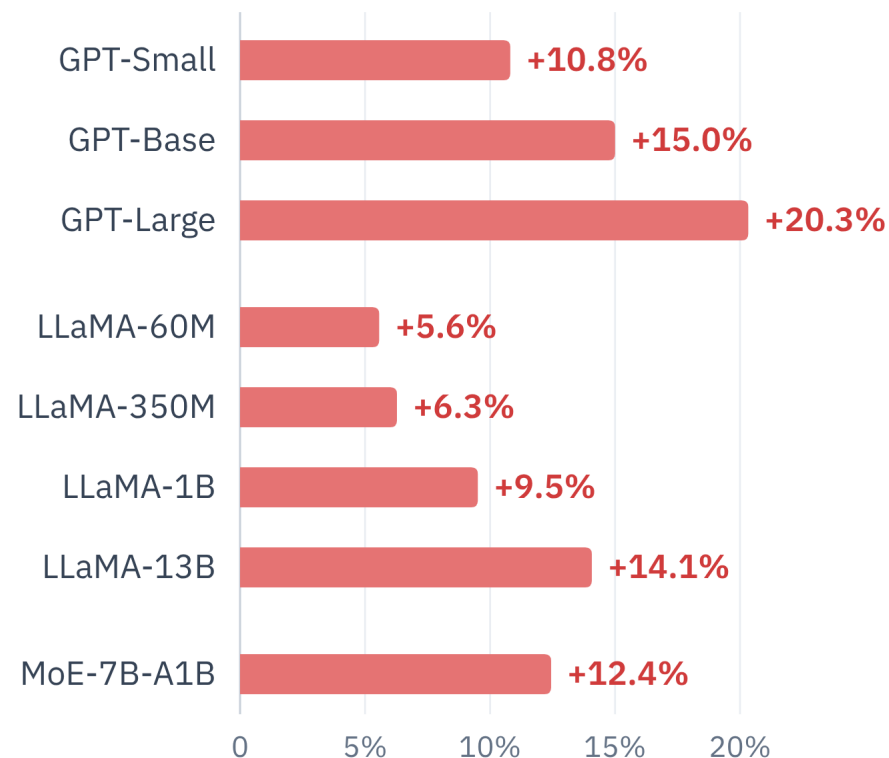
$(a, b, c) = (3.4445, -4.7750, 2.0315)$, repeated $K = 5$ times

- **3 matrix multiplications** on the full matrix per step
- Weights are **sharded** in distributed training, so each step needs extra compute or communication
- Fewer steps would be cheaper, but **quality drops**

Can we use fewer steps *and* get a better optimizer?

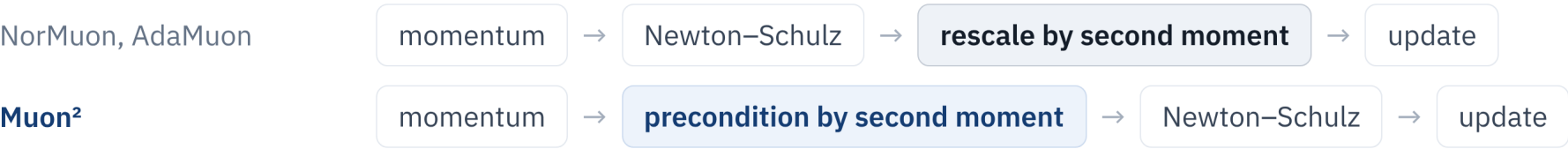
Muon with 3 instead of 5 steps

increase in validation perplexity

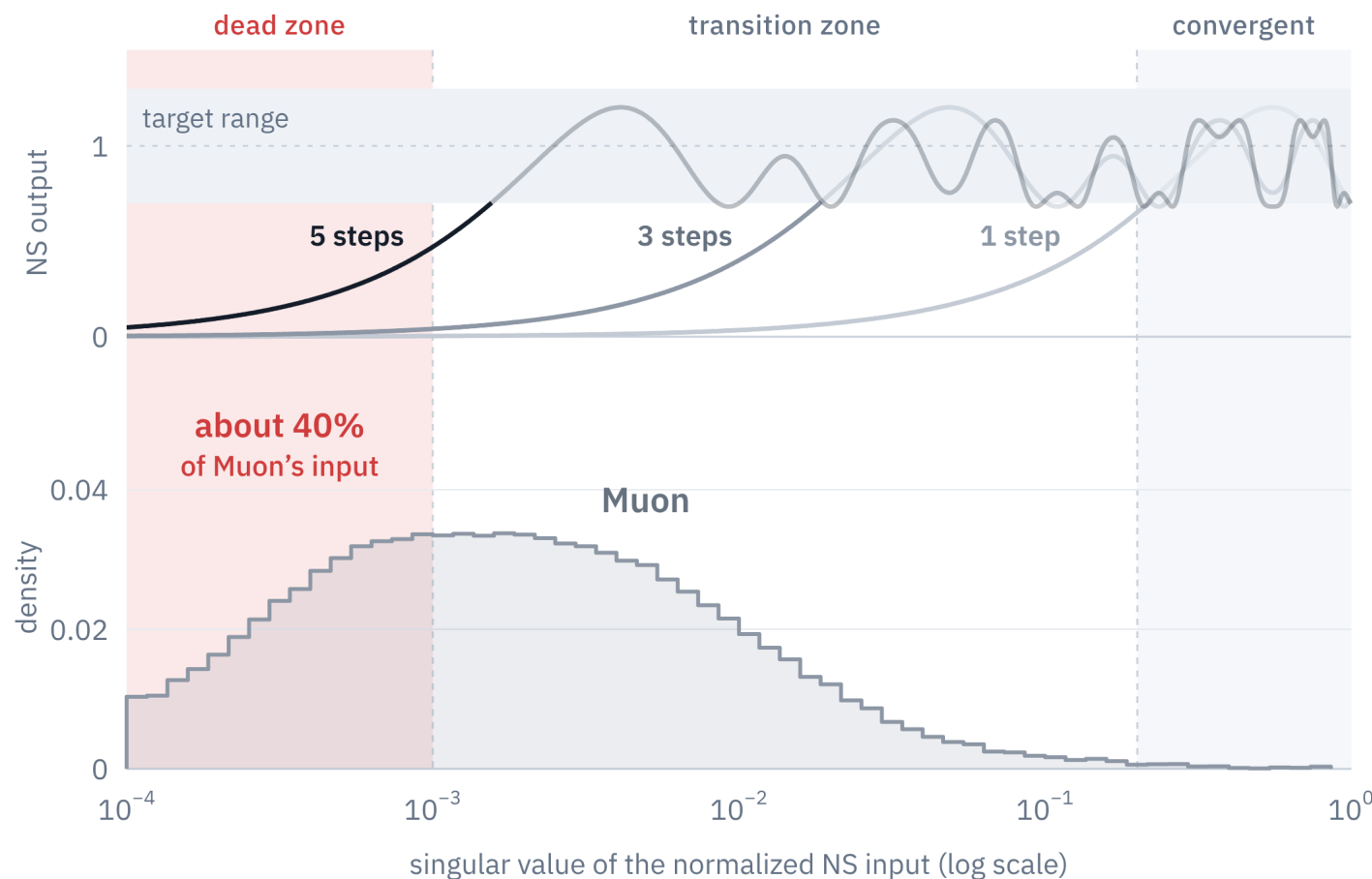


Existing Muon variants improve one side, not both

Method	What it changes	Targets efficiency fewer Newton–Schulz steps	Targets effectiveness better model performance
Turbo-Muon Boissin et al., 2025	preconditions the NS input	● 5 → 4 steps	○
PolarExpress Amsel et al., 2026	optimal polynomial for each NS step	○	●
NorMuon Li et al., 2025	second moment on the NS output	○	●
AdaMuon Si et al., 2025	second moment on the NS output	○	●
Muon² (ours)	second moment on the NS input	● 5 → 3 steps	●



Many singular values are too small for Newton–Schulz



- Each step maps every singular value through

$$\phi(\sigma) = a\sigma + b\sigma^3 + c\sigma^5$$
- Small values grow only about 3.4× per step, so below about 10^{-3} five steps are not enough: the **dead zone**
- Early in training, **about 40%** of Muon's input sits there (LLaMA-60M)

The bottleneck is the **ill-conditioned input**.

Can we fix the input instead?

Muon²: precondition the momentum, then orthogonalize

$$1 \quad \mathbf{G}_t = \nabla_{\mathbf{W}} \mathcal{L}(\mathbf{W}_t) \quad \text{gradient}$$

$$2 \quad \mathbf{M}_t = \beta_1 \mathbf{M}_{t-1} + (1 - \beta_1) \mathbf{G}_t \quad \text{momentum}$$

$$3 \quad \mathbf{V}_t = \beta_2 \mathbf{V}_{t-1} + (1 - \beta_2) \mathbf{G}_t \odot \mathbf{G}_t \quad \text{added: second moment}$$

$$4 \quad \widetilde{\mathbf{M}}_t = \mathbf{M}_t \oslash (\sqrt{\mathbf{V}_t} + \epsilon) \quad \text{added: precondition}$$

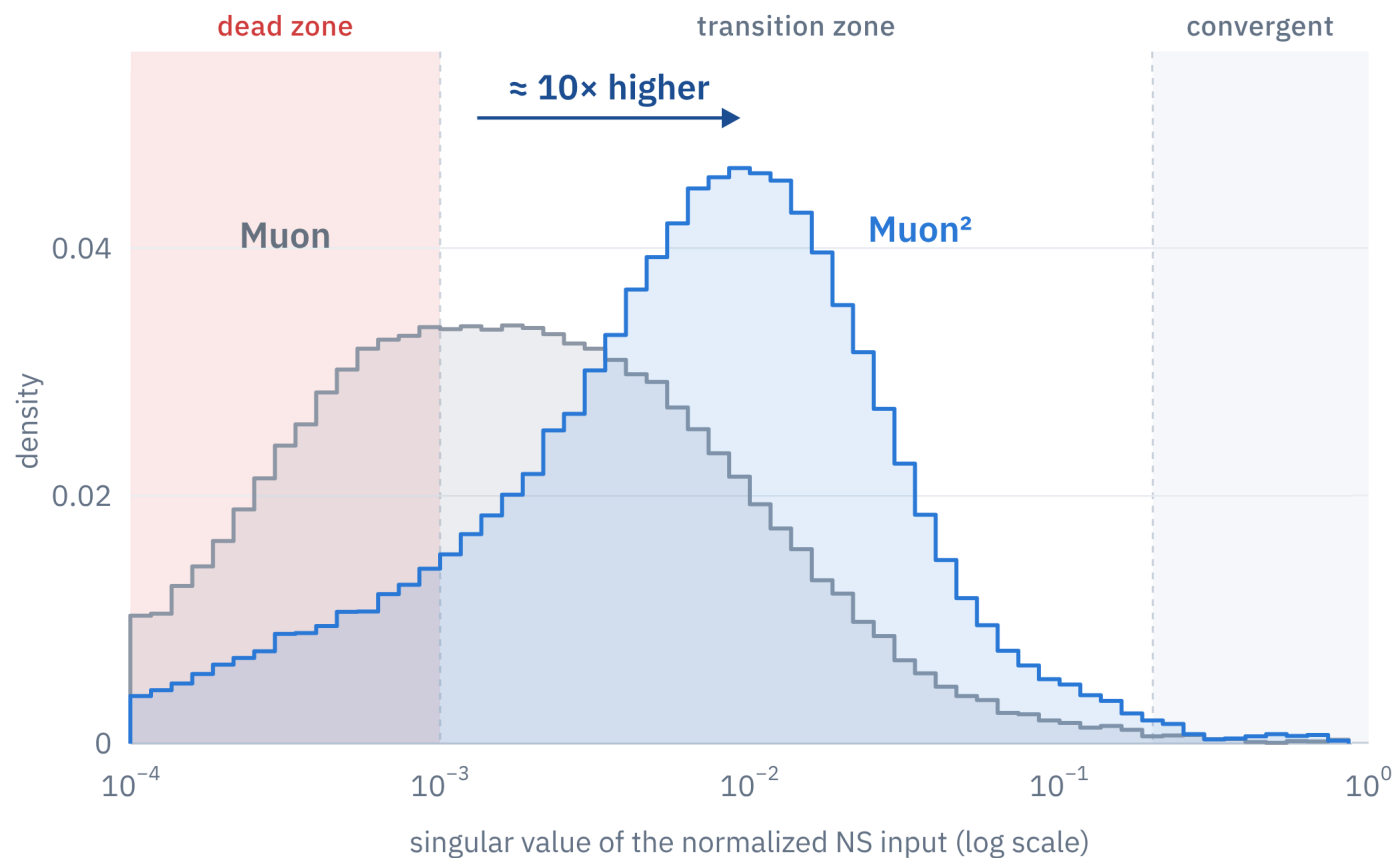
$$5 \quad \mathbf{O}_t = \text{NewtonSchulz}(\widetilde{\mathbf{M}}_t, K) \quad \text{orthogonalize}$$

$$6 \quad \mathbf{W}_{t+1} = \mathbf{W}_t - \eta \sqrt{m/n} \mathbf{O}_t \quad \text{update}$$

- Adam-style second moment, applied **before** Newton–Schulz. **Everything else is Muon.**
- It conditions the NS **input**, not the step size: the update stays orthogonalized
- Cost: a few element-wise operations. Memory: one extra matrix, which Muon²-F^{*} removes

* Muon²-F: memory-efficient version that keeps only row and column statistics of the second moment (Adafactor-style); see slide 10.

Preconditioning moves the spectrum out of the dead zone



Dead-zone fraction ↓



Effective rank ↑



A random-matrix view: dividing by the scale removes the dead zone

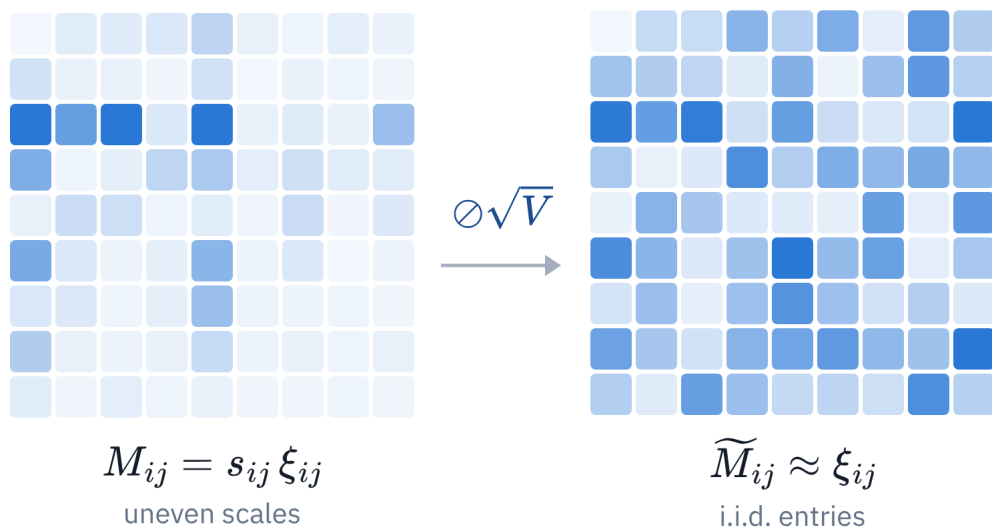


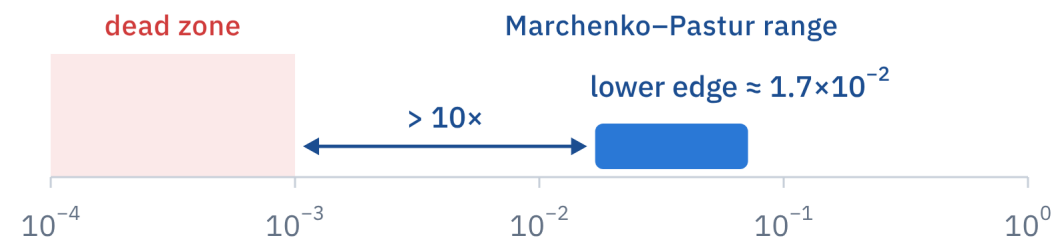
Illustration. ξ_{ij} : i.i.d., zero mean, unit variance. $\sqrt{V}$ estimates the scale s_{ij} .

MARCHENKO-PASTUR

Singular values of $\tilde{M}/\sqrt{n}$ fill
 $[1 - \sqrt{\lambda}, 1 + \sqrt{\lambda}]$, $\lambda = m/n$

For rectangular layers ($\lambda < 1$) this is **bounded away from zero**: the dead zone vanishes for any scale profile.

LLaMA-60M MLP (512×1344 , $\lambda \approx 0.38$), after normalization:



Also holds with a signal (a nonzero mean) in the momentum: a low-rank signal of bounded strength only lowers the edge by a constant factor.

Measured by direction, Muon² needs fewer steps

Muon's own coefficients map singular values only to about **[0.7, 1.3]**, and Muon trains well with them:

10×

larger orthogonality error $\|\mathbf{Q}^\top \mathbf{Q} - \mathbf{I}\|$ than coefficients that target exactly 1

0.98

cosine with the exact polar factor: the direction is almost exact

The learning rate absorbs any global scale, so we measure the **direction**:

$$\cos(\mathbf{Q}, \mathbf{Q}_\star) = \frac{\langle \mathbf{Q}, \mathbf{Q}_\star \rangle_F}{\|\mathbf{Q}\|_F \|\mathbf{Q}_\star\|_F}$$

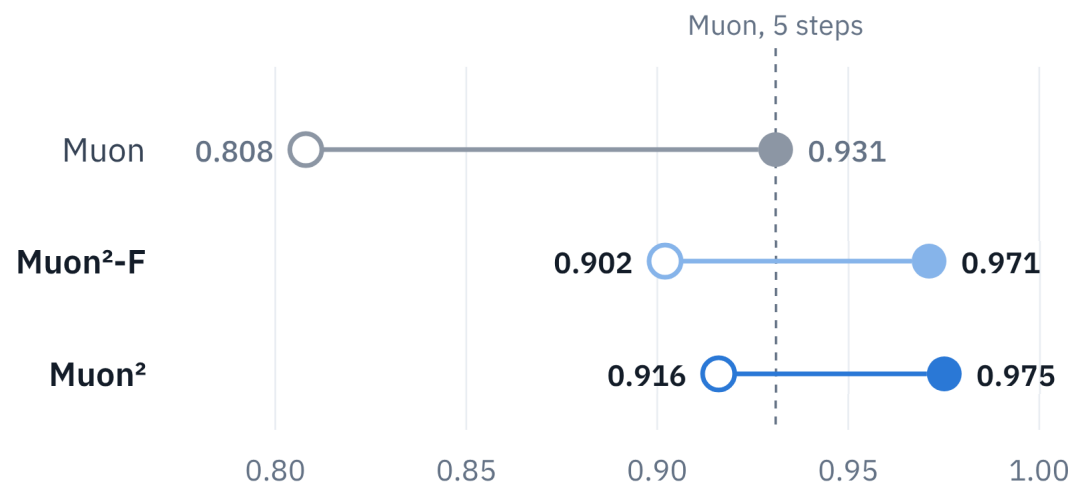
$\mathbf{Q}_\star = \mathbf{U}\mathbf{V}^\top$: the exact polar factor

10× and 0.98: synthetic check in the paper (App. B), singular values on a uniform grid in [0, 1], 5 steps.

Alignment with the exact orthogonalized update

cosine similarity, LLaMA-60M, end of training

○ 3 steps ● 5 steps



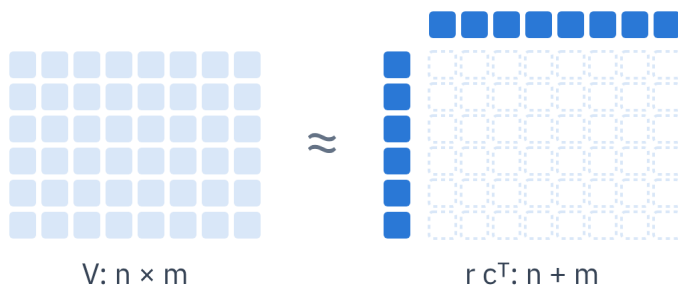
GPT-2 Small agrees: Muon² with 3 steps reaches 0.84, Muon with 5 steps 0.86.

Muon²-F keeps the memory of Muon

$\mathbf{V}$ only conditions the NS input, so an approximation should be enough.

Adafactor-style: keep moving averages of the **row and column sums** of $\mathbf{G}_t \odot \mathbf{G}_t$, then

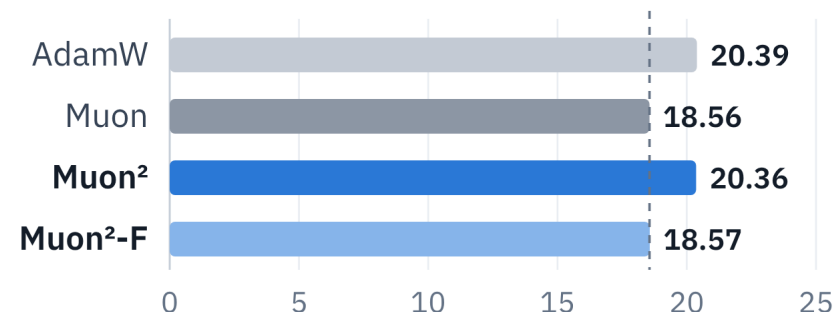
$$\hat{\mathbf{V}}_t = \frac{\mathbf{r}_t \mathbf{c}_t^\top}{\mathbf{1}^\top \mathbf{r}_t}$$



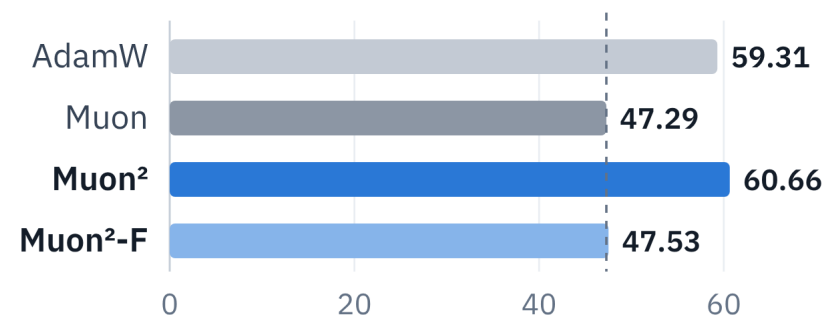
Peak training memory, GB

one H100, bf16, sequence length 4096, micro batch 1; dashed line: Muon

LLAMA - 1B

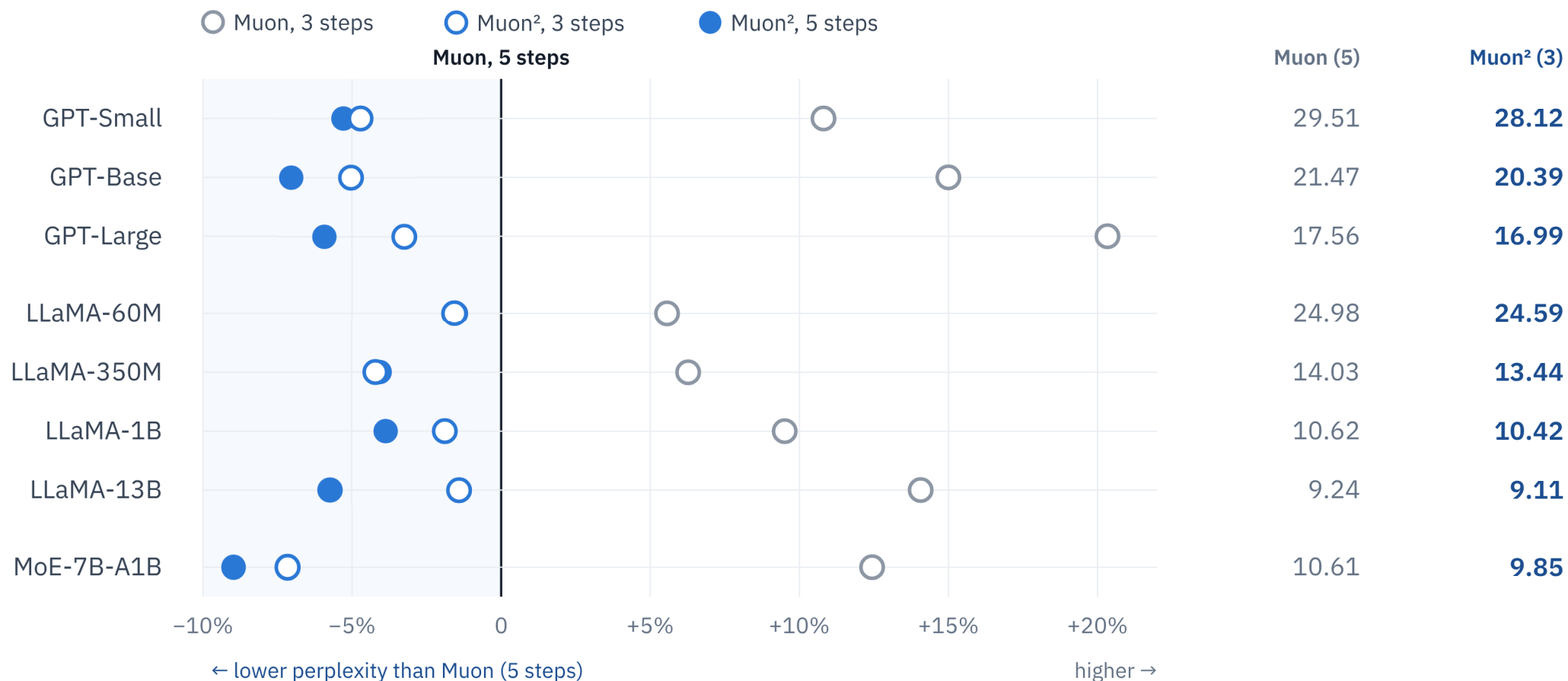


LLAMA - 7B



With 3 steps, Muon² beats Muon with 5 on every model

GPT Small, Base, Large on FineWeb · LLaMA 60M to 13B on C4 · MoE 7B with 1B active · learning rate swept for both optimizers up to 1B



No tuned Muon variant matches Muon², even with more steps

validation perplexity	GPT-Small		GPT-Base	
	3 steps	5 steps	3 steps	5 steps
Muon²	28.12	27.95	20.39	19.96
Muon	32.70	29.51	24.69	21.47
PolarExpress polar approximation	30.01	29.42	22.74	21.16
Turbo-Muon polar approximation	29.70	29.66	23.46	21.93
NorMuon update rule	30.35	28.40	23.33	21.27
AdaMuon update rule	31.20	29.30	26.07	22.42

Muon² with **3 steps** is better than **every variant with 5 steps**.

GPT-Small: 28.12 vs. 28.40 at best

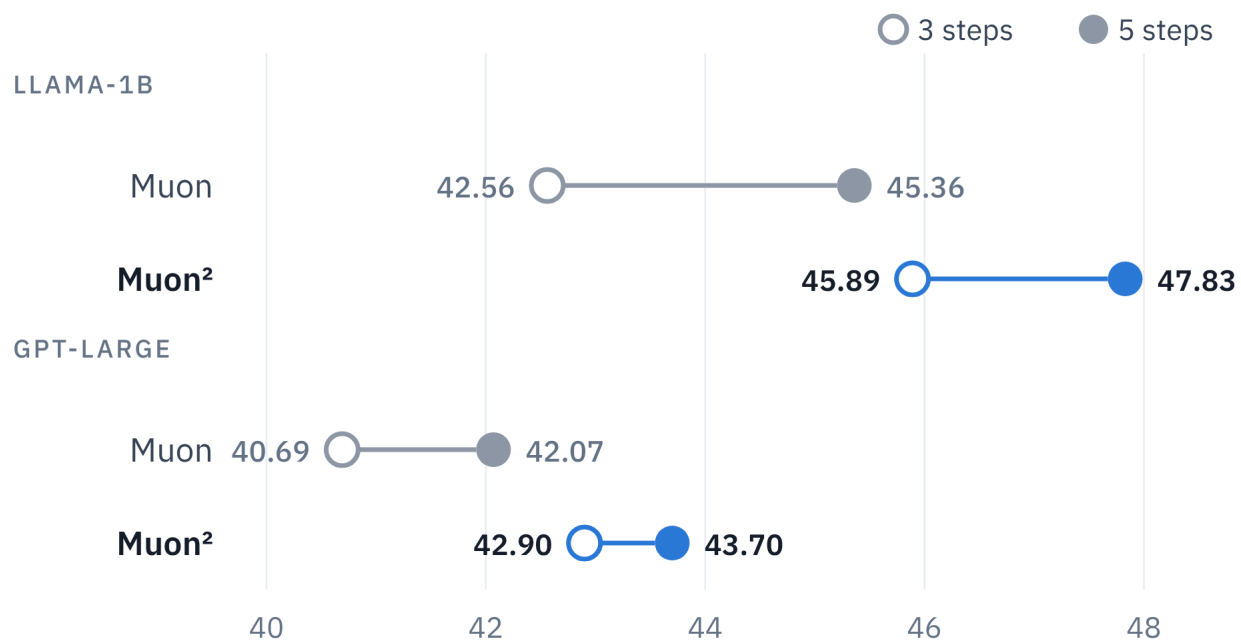
GPT-Base: 20.39 vs. 21.16 at best

- Polar methods lose less than Muon at 3 steps, but stay behind
- Update-rule methods do not allow fewer steps, and at 5 steps only NorMuon beats Muon on both models
- All methods tuned over learning rate

The gains carry over to downstream tasks

Average accuracy over eight tasks, %

ARC-c, ARC-e, OBQA, HellaSwag, PIQA, WinoGrande, LAMBADA, MMLU



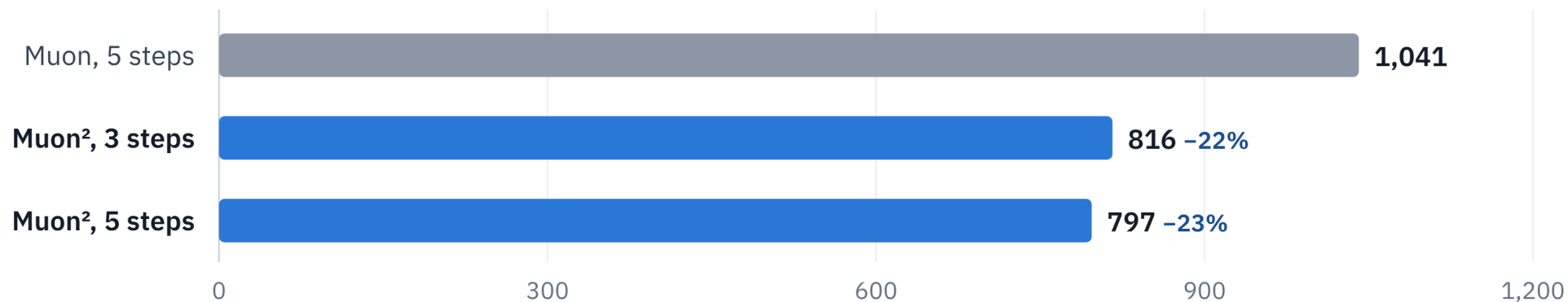
Muon² with 3 steps is ahead of **Muon with 5 steps** on both models.

At this scale ARC-c, WinoGrande and MMLU are near chance zero-shot, so every model is fine-tuned on these three with the same protocol before evaluation. The other five tasks are zero-shot.

Same loss as Muon with 23% fewer GPU-hours

GPU-hours to reach a loss of 2.36 on LLaMA-1B

2.36 is the final loss of Muon with 5 steps; fewer is better



	Muon, 5 steps	Muon², 3 steps	Muon², 5 steps
Training steps	4,796	3,862	3,632
Time per step	3,053 ms	2,971 ms	3,085 ms

64 nodes × 4 GPUs. The element-wise scaling adds almost no time per step, so fewer steps means less time.

Limitations and future work

01

Frontier scale

We trained up to 13B parameters on an academic budget, already larger than most related work. Frontier-scale runs need industrial resources, and we are open to collaboration.

02

Memory-efficient designs

Muon²-F is one way to compress the second moment. Other designs may balance optimization quality and optimizer-state cost better.

03

Distributed training

Every Newton–Schulz step needs the full matrix. We want to measure the savings of fewer steps in large sharded training.

MORE MUON WORK FROM OUR GROUP

TEON

Muon+

MuonQ COLM 2026

Thank you! *Questions?*

TAKEAWAYS

- 1 Muon's Newton–Schulz struggles because its **input is ill-conditioned**.
- 2 **Muon²** divides the momentum by an Adam-style second moment **before** orthogonalizing. One extra line.
- 3 Lower perplexity with **40% fewer NS steps** up to 13B, and Muon's final loss with **23% fewer GPU-hours**.



PROJECT PAGE
alvin-zyl.github.io/muon2

PAPER
arXiv 2604.09967

CONTACT
ziyueliu@ucsb.edu